Surya Thirukonda

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Education

Texas A&M University College Station, TX
Bachelor of Science in Computer Science; GPA: 4.0 *May 2028*

Experience

Software Engineering Intern, Quantitative Systems June 2026 – Present
DeKalb Capital Management *Remote*

- Built portfolio analytics and stock screening on a Dockerized FastAPI dashboard serving 100+ equities, integrating yfinance and the Interactive Brokers API for position sync and time-weighted returns.
- Made brokerage sync idempotent and retry-safe, eliminating duplicate position writes and stale reads; added validation, structured logging, rate limiting, and health checks.

Founding Software Engineer (Part-time) Feb. 2026 – Present
Revio *Remote*

- Launched a manuscript-formatting beta supporting 40+ journal profiles and 200+ end-to-end conversions, architecting a Dockerized FastAPI and React/Vite platform through production.
- Built async, schema-validated document processing on Gotenberg DOCX/PDF pipelines with Supabase and OAuth, containerized across Railway and Vercel.

Contract Software Engineer Feb. 2026 – June 2026
Mai Shan Yun (WaiterDock) *College Station, TX*

- Won the Mai Shan Yun sponsor challenge at TAMUhack (40+ teams) and converted the 24-hour prototype into a four-month paid development contract.
- Synchronized order state across admin, waiter, kitchen, and customer roles with RBAC, Stripe payments, REST APIs, and WebSocket updates.

Research & Projects

Undergraduate Researcher (REU) June 2026 – Present
Portland State University *Remote*

- Investigated whether regime-aware modeling improves short-horizon Treasury yield forecasts, cutting 20-day curve RMSE 3.4% versus a random walk (28.03 to 27.06 bp over 1,149 rolling origins).
- Generated 119,000+ out-of-sample forecasts with per-origin retraining, under an evaluation design excluding unrealized targets from every training window.

Software Development Engineer, VEST Project Jan. 2026 – May 2026
Turtle Robotics, Texas A&M *College Station, TX*

- Streamed 4 biometric channels at 12 samples/second for a wearable canine health vest via an ESP32-to-Supabase-to-React Native pipeline.
- Designed the Supabase schema and REST endpoints serving live readings and historical trends to the React Native interface.

RateSignal | *Node.js, Express, PostgreSQL, Prisma, Redis, Docker* June 2026 – July 2026

- Built and deployed a Treasury-yield forecasting platform exposing 1,000+ forecast records across 11 maturities, 7 models, and 3 horizons via authenticated REST APIs.
- Secured the API with JWT and Argon2 hashing, Zod validation, Redis-backed rate limiting, OpenAPI docs, and GitHub Actions CI.

Physio | *MediaPipe, FastAPI, WebSockets, Raspberry Pi, ElevenLabs* May 2026

- Placed 2nd overall at HackHCC 2026 (40+ teams) with a voice-first physical-therapy assistant turning at-home rehab into structured motion data.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, R
Backend/Web: FastAPI, Node.js, Express, React, REST APIs, WebSockets, Prisma, SQLAlchemy, JWT, OAuth
Infrastructure: PostgreSQL, QuestDB, Redis, Supabase, Docker, GitHub Actions, AWS, Railway, Cloudflare, Git, Linux
Data/ML: pandas, NumPy, scikit-learn, statsmodels, XGBoost, SARIMAX, VAR, Dynamic Nelson-Siegel, HMM/HSMM
Certification: AWS Certified Cloud Practitioner